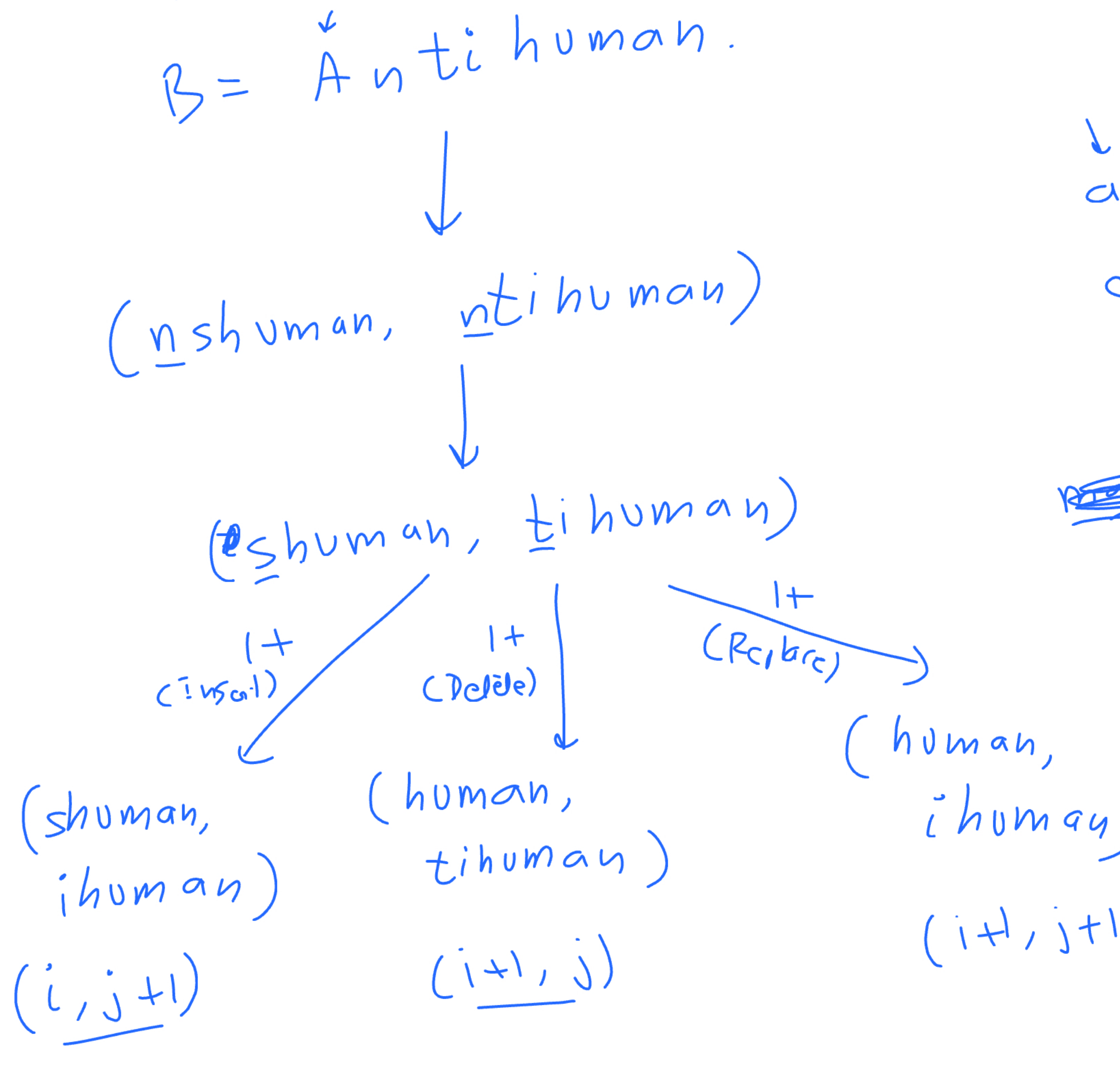
0-1 Knapsack (Bottom-Up)

dp[i][j] = max value that can be generated using items[0 ... i] with capacity = j

---

A = Anshuman
B = Antihuman

ab
cb


(nshuman, ntihuman)
(shuman, tihuman)
1+ (insert)
1+ (Delete)
1+ (Replace)
(shuman, ihuman)
(i, j+1)
(human, tihuman)
(i+1, j)
(human, ihuman)
(i+1, j+1).

ED(i, j) = ED(i+1, j+1) if A[i] == B[j].

1+ min(ED(i, j+1), ED(i+1, j), ED(i+1, j+1)) otherwise.

abac
bbac

1+ ED(bac, bac) = 1

A = "" B = "abc"

B[j ... m-1].

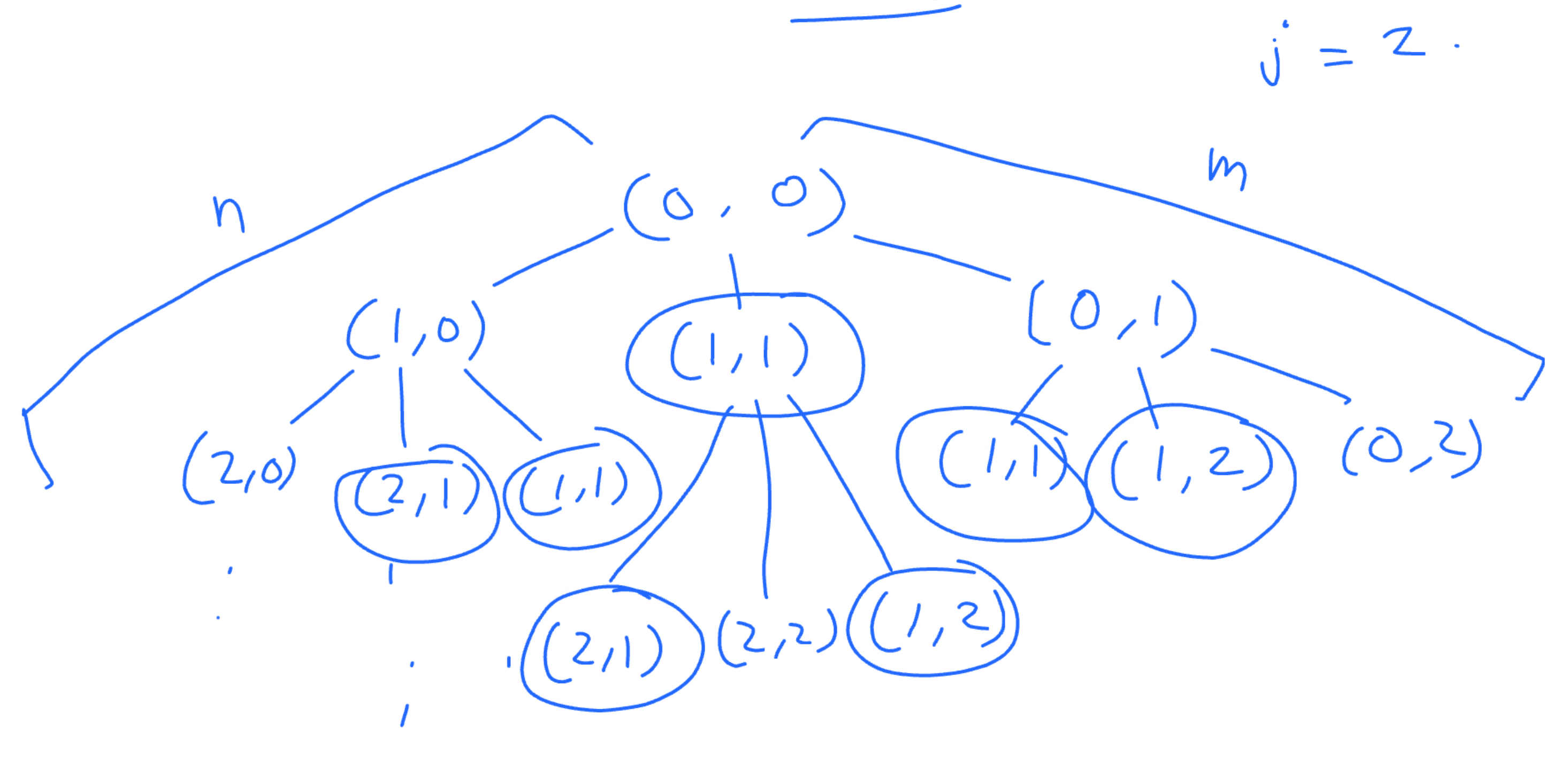
A[i ... n-1], B[j ... m-1].

0	1	2	3	4	5	6
a	b	c	d	e	f	g

~~m=5~~.

m-j

m = 7
j = 2.


n
m
(0, 0)
(1,0)
(1,1)
(0,1)
(2,0)
(2,1)
(1,1)
(2,1)
(2,2)
(1,2)
(1,1)
(1,2)
(0,2)

H = max(n, m).

TC = $O(3^{\max(n,m)})$.
AS = $O(\max(n,m))$.

n = 7.

0	1	2	3	4	5	6	7
0	1	1	2	3	5	8	13

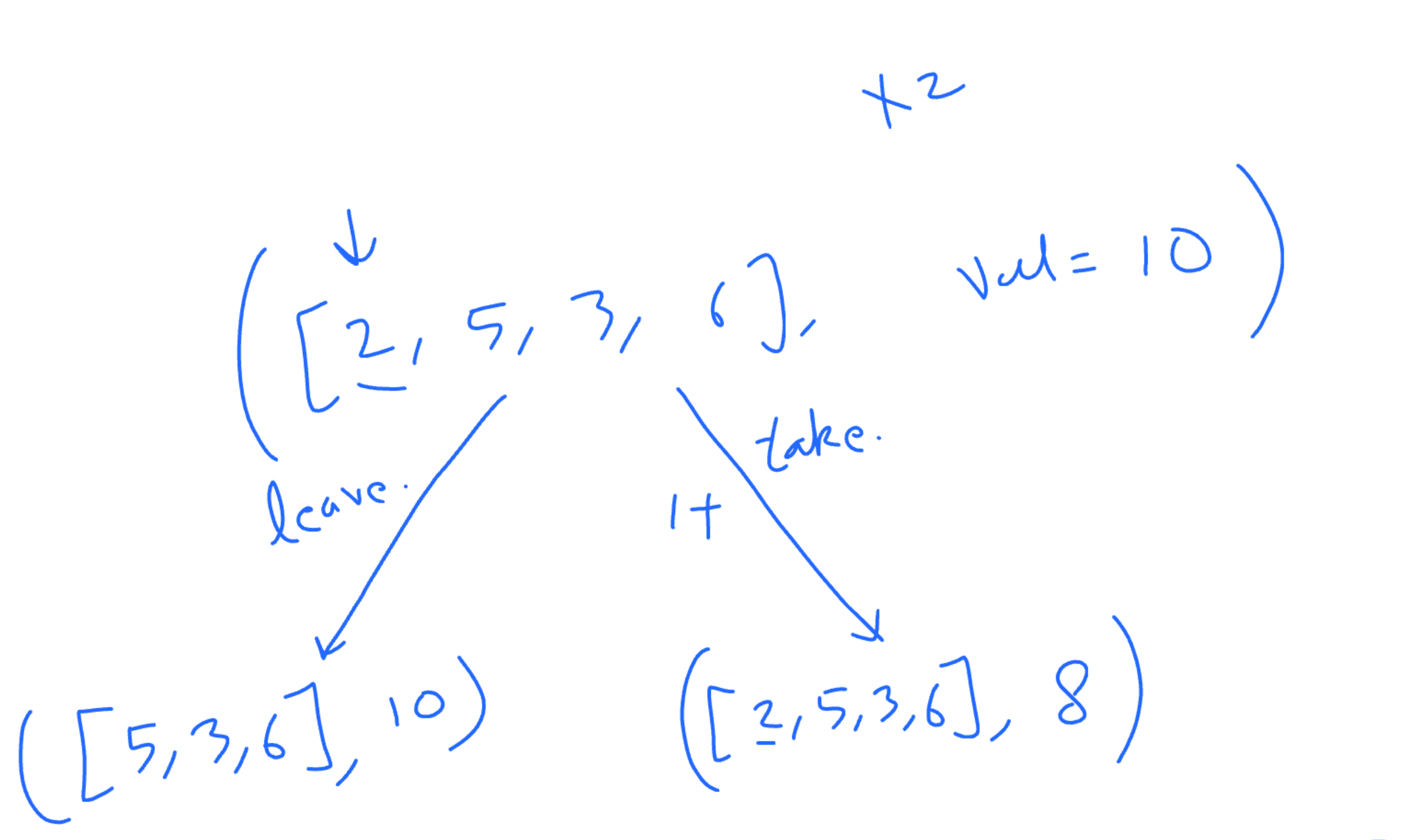
Min Coins to Make a Change.

[2, 5, 3, 6]. value = 10.

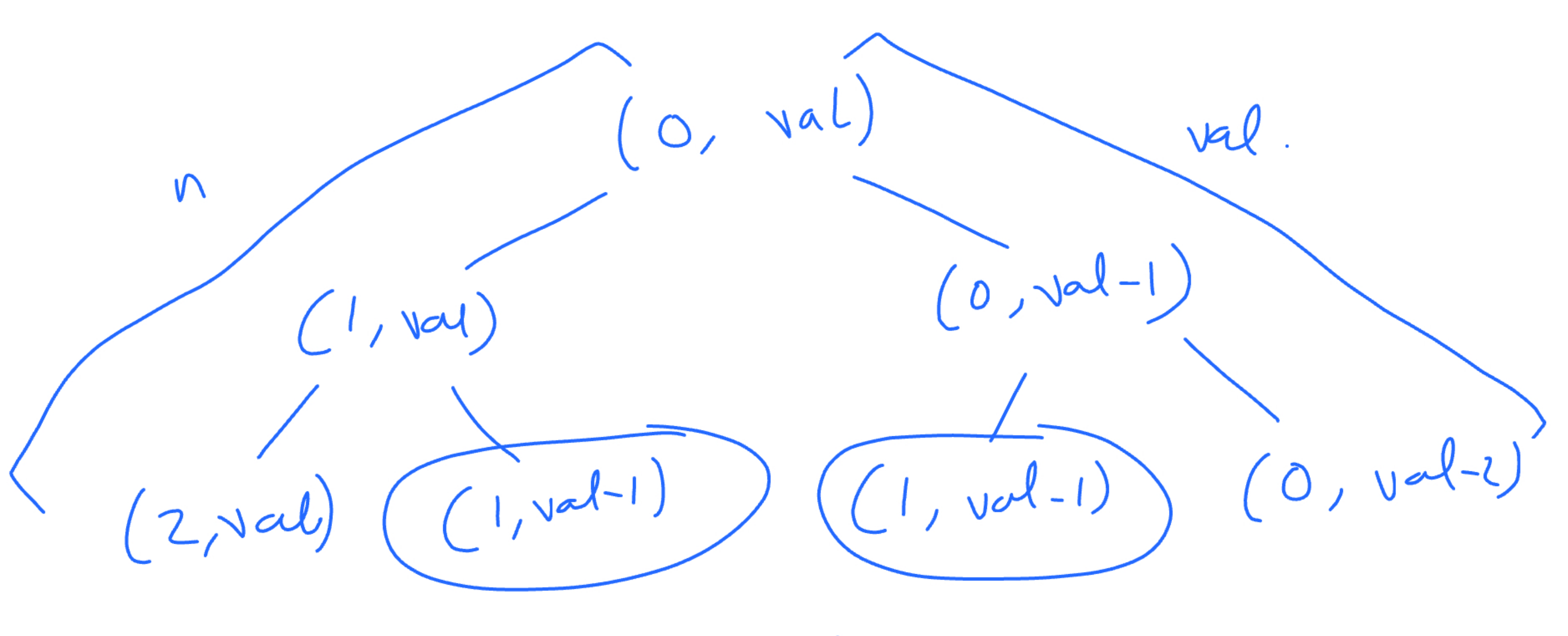
(2, 3, 5, 6)

~~10~~ ~~4~~ 1

×2


([2, 5, 3, 6], val = 10)
leave.
1+ take.
([5,3,6], 10)
([2,5,3,6], 8)

(i, val) leave → (i+1, val)
take → 1+(i, val - coin[i])
} min


n
val.
(0, val)
(1, val)
(0, val-1)
(2, val)
(1, val-1)
(1, val-1)
(0, val-2)

H = max(n, val)

TC = $O(2^{\max(n, val)})$.
AS = $O(\max(n, val))$

[~~5~~, ~~3~~, ~~7~~, ~~10~~].

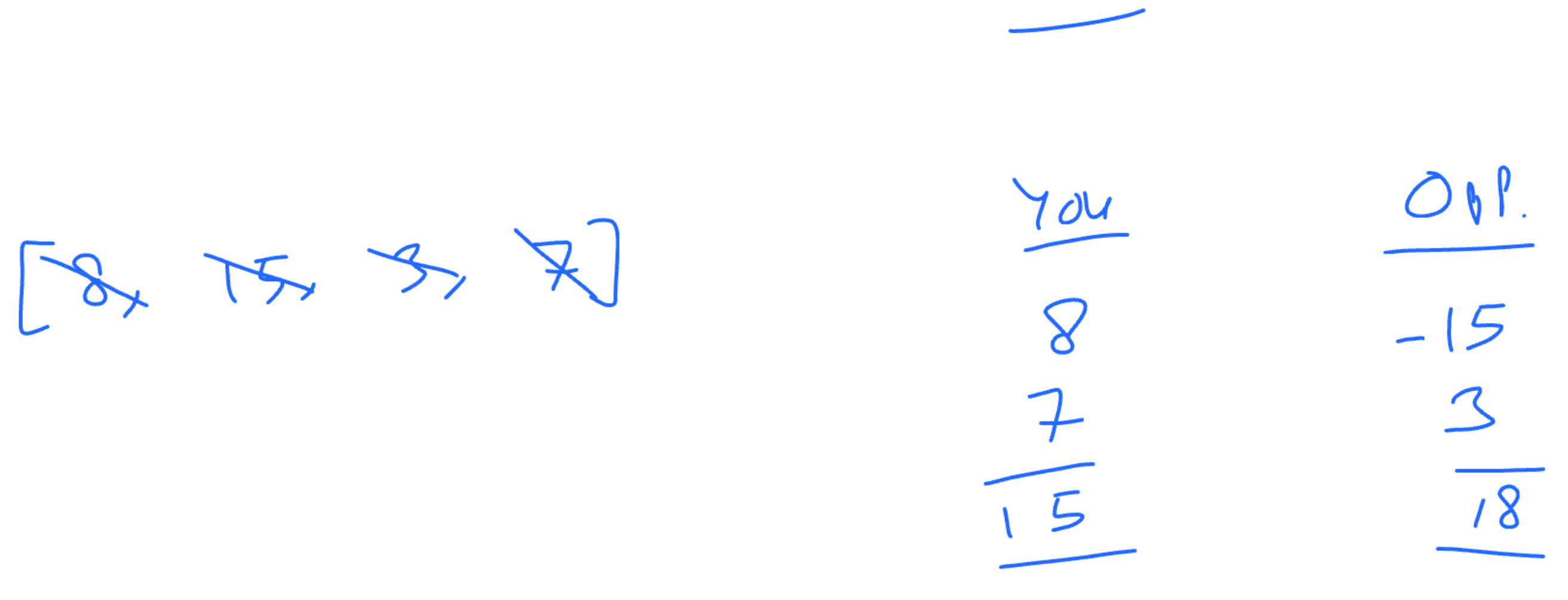
You	Opp.
10	7
5	3
15	10

[~~8~~, ~~15~~, ~~3~~, ~~7~~]

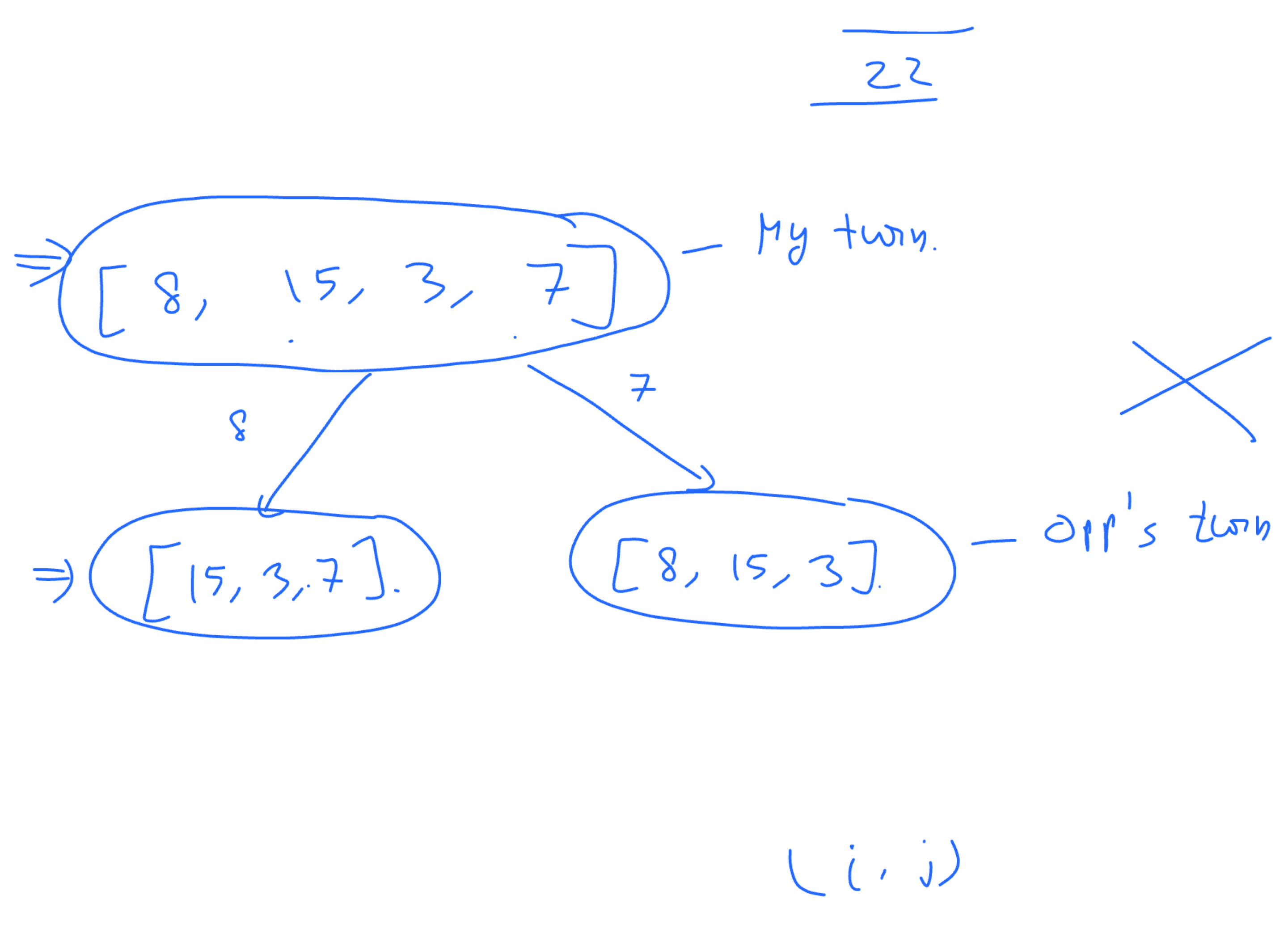
You	Opp.
8	-15
7	3
15	18

[~~8~~, ~~15~~, ~~3~~, ~~7~~]

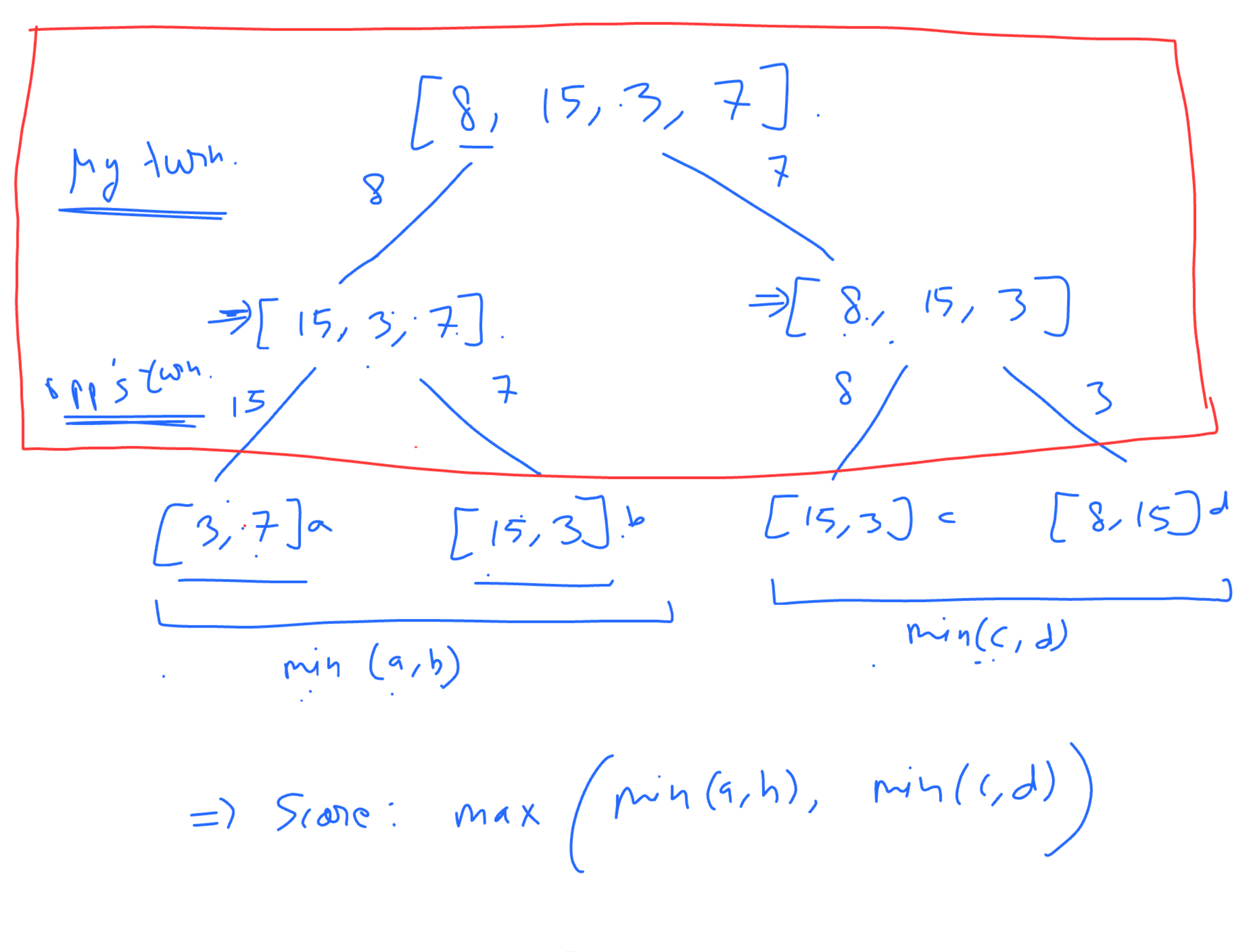
You	Opp.
7	8
15	3
22	


[8, 15, 3, 7]
My turn.
8
7
[15, 3, 7].
[8, 15, 3]
Opp's turn

(i, j)


My turn.
[8, 15, 3, 7].
8
7
Opp's turn.
[15, 3, 7].
[8, 15, 3]
15
7
8
3
[3, 7] a
[15, 3] b
[15, 3] c
[8, 15] d
min (a, b)
min(c, d)

=> Score: max (min(a, b), min(c, d))


My turn.
(i, j)
i
j
Opp's Turn
(i+1, j)
(i, j-1)
i+1
j
i
j-1
(i+2, j)
(i+1, j-1)
(i+1, j-1)
(i, j-2)

max(min((i+2, j), (i+1, j-1)), min((i+1, j-1), (i, j-2)))